



SAP Commerce Cloud

Cloud Readiness Check

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Document Revision History

Date	Revision	Author	Summary of Changes
October 18, 2022	1.0	Serge Akoa	Initial Version
October 26, 2022	2.0	Serge Akoa	Final Version

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1 INTRODUCTION

This report contains the findings from the Cloud Readiness Check (CRC), which aims to reduce risks when deploying your solution for the first time on SAP Commerce Cloud. A review of your solution has been executed to highlight potential issues and make recommendations on areas of your solution that may experience issues on or after go-live.

The CRC is intended as a check to ensure your solution is reviewed for many of the issues commonly found prior to going live in your production environment. It does **not** include system nor application fixes because of the review performed. Any actions recommended as a result of the CRC, will have to be reviewed and completed by CRC Test for One Strike and/or SAP CX.

If an in-depth review is required, we encourage you to further explore how review and validation services offered by SAP can help to further boost the quality of your project. Please contact your Customer Engagement Manager for more details.

1.1.1 Activities

The following tasks were planned and executed as of October 24, 2022 and presented to Customer.

Area	Description
Project Setup	Review of your configuration to ensure it meets the standards for deploying on Commerce Cloud
Source Code Analysis	A high-level static source code analysis of the custom codebase using SonarQube; Type system validation;
Integrations	Brief inspection of your integrations for compatibility and meets recommended design patterns. Running/testing of the integrations is out of scope.
Performance Testing Approach	A high-level evaluation to look for possible performance problems that will require further investigation before go-live
Data Maintenance	Review of the data being generated: what data is live, what data is being archived, and what data can be removed to keep the system running at peak performance
Application Security Review	Review of security practices. Ensuring best practices are being followed to guarantee a secure system

Each chapter in this report contains detailed information on improvements, further issues and recommendations to be investigated in each area. These activities are in line with the recommended practices collected in our methodology. Further information can be found in the [official documentation](#) or on <https://www.sap.com/cxworks/>.

2 SUMMARY OF RECOMMENDATIONS

The following table provides a list of all recommendations found during the execution of the Cloud Readiness Check. Each recommendation has a reference to a corresponding numbered section with the full details.

- **Blocker** – Needs to be fixed before go-live. These are very likely to result in a negative impact on your solution at go-live.
- **High** – Should be fixed by go-live. If not possible, plans should be included to fix as part of fast-follow release as there is still high chance of an issue occurring.
- **Medium** – Does not need to be fixed prior to go-live. Should create plans to fix within 3-5 months of going live.
- **Low** - Does not need to be fixed prior to go-live. Should create plans to fix within 3-12 months of going live, as it may have a detrimental impact the longer your solution runs.

Issue	Recommendation	Section	Priority
Commerce Version	You are running a version of SAP Commerce Cloud that will soon reach End of Mainstream Maintenance. When a version of SAP Commerce Cloud enters the End of Mainstream Maintenance (EoMM), SAP no longer releases any patches for it, including corrections or security updates. This is defined in the SAP Commerce Cloud Supplemental Terms and Conditions .	3.1	BLOCKER
SOLR Version	Upgrade your Solr to an official supported version, using the latest patch release	3.1	BLOCKER
Performance Testing	We recommend performing performance tests for the SSO Client to ensure that the integration can cope with the load coming from commerce.	6.2	HIGH
Missing password encoding	Some passwords were found to not be encoded. Review password encoding configuration.	8	HIGH
Default users	We found some issues, please refer to the table below. However, since we did the check on S1, we highly recommend conducting the test in production to ensure that only necessary default users are used but also the corresponding encryption method is being used.	8	HIGH
Bugs & Vulnerabilities	8 blockers found. Address Blocker bugs before go-live as well as the 5 security hotspots (even though after our analysis we did not find any severe issues). Regarding security, Sonar detected Hard-coded IP Addresses which are generally not a good idea because doing so, we introduce Infrastructure concerns into application concerns. If the IP were to change, the health check would no longer work. Please review the piece of code (see below):	4.1	HIGH
Type System	We identified some issues with custom types you have defined for your solution. Please see the section for individual issues identified	4.3	HIGH
Integrations	A lot of integration points are part of the landscape. However, if the different systems are part of the on-premises landscape, we highly recommend considering introducing VPN to secure connections to the hosted systems.	5.1	HIGH

Critical Business Processes	Please ensure that the cronjobs are running on the background processing node(see attachment).	4.6	HIGH
Cache Settings	The cache setting is currently configured as static with default values. SAP Commerce Cloud leverages the existing Region Cache of SAP Commerce. However, since the SAP Commerce Cloud build process controls the Java heap size, which can be different per environment, it is required to configure cache regions in a flexible way instead of having fixed values inside. The maximum number of entries for a given Cache Region is set using SPEL. The Max Memory from the JVM is retrieved and used	3.1	HIGH
Search Restrictions	There are some custom search restrictions found that might lead to some issues, please review the detailed list.	4.5	MED
Manifest	Solr Extension is still part of the deployment package. We highly recommend removing it from the deployment package.	3.2	MED
Large Tables	Your solution contains some large tables, which is usually an indication that cleanup jobs aren't in place.	7.1	MED
Cleaning-up Jobs	SAP recommends declaring and activate all clean up cronjob necessary to keep database platform in good shape. Beware: the information here can be interesting for an audit trail. Make sure to backup the database and only cleanup if there is no need for these entries anymore.	7.2	MED
CronJob Cleanup	Configure and activate CleanupCronJobStrategy based on your audit requirements to ensure you are clearing out cronjobs to not fill up your database.	7.2	MED
Redirects	No redirection sets were defined. We commonly see redirects for a Commerce Cloud solution. They can be defined for an endpoint, enabling vanity & marketing URLs and site migrations. Confirm it is ok to not have any redirect sets defined. See documentation	3.3	MED
Cloud Configuration	Add a maintenance page especially for the production environment	3.3	MED
Certificates	Configured but some are outdated and need to be removed. create an SAP Ticket to the support team	3.3	LOW

3 PROJECT SETUP

3.1 Compatibility & Startup

SAP Commerce Cloud sets some default values:

- for cache, 800k

Configuration Properties

apply all reset all

Show unsaved only: ☐ OFF

Show 10 entries Search: cache.en

Key	Value
New key...	New value... add
regioncache.entityregion.evictionpolicy	LRU
regioncache.entityregion.size	800000

Showing 1 to 2 of 2 entries (filtered from 2,958 total entries) Previous 1 Next

But the default RAM size is 6 GB. A good rule of thumb is to have a maximum of 40k cache entries per 500MB RAM

(https://help.sap.com/docs/SAP_COMMERCE/d0224eca81e249cb821f2cdf45a82ace/8c2780e686691014a134cc8fd1091aa3.html?version=2011&locale=en-US#loio8c2780e686691014a134cc8fd1091aa3_performancetuningoverview-import)

Which means 80k cache corresponds to 1GB of RAM and 480k to 6 GB RAM.

By this rule, an 800k cache would need 10 GB RAM which does not corresponds to the default setting.

Property	Further Reading	Issues Found
The current SAP Commerce version is supported by Commerce Cloud. It is recommended to be on the latest version of SAP Commerce to be able to take advantage of all the features.	https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/1be46286b36a4aa48205be5a96240672/5c519f566fb744efb42d49543bccb2b7.html?locale=en-US&version=v2105	You are running a version of SAP Commerce Cloud that will soon reach End of Mainstream Maintenance. When a version of SAP Commerce Cloud enters the End of Mainstream Maintenance (EoMM), SAP no longer releases any patches for it, including corrections or security updates. This is defined in the SAP Commerce Cloud Supplemental Terms and Conditions .
The current SAP Commerce SOLR setup is not supported by SAP Commerce Cloud. Used version is only under legacy compatibility.	Third Party Compatibility	Yes, version used 1905 is no longer maintained by SAP and hence Solr must be upgraded as well.
The cache setting is currently configured as static with default values. SAP Commerce Cloud leverages the existing Region Cache of SAP Commerce. However, since the SAP Commerce Cloud build process controls the Java heap size, which can be different per environment, it is required to	https://www.sap.com/cxworks/article/2589633403/sap-commerce-cloud-architecture#SAPCommerceCloudArchitecture-Caching	Yes

configure cache regions in a flexible way instead of having fixed values inside. The maximum number of entries for a given Cache Region is set using SPEL. The Max Memory from the JVM is retrieved and used to calculate the number of entities allowed in the cache region.

3.2 Manifest File

The manifest is a key component in telling the build automation tooling in Commerce Cloud how to build your solution. We have reviewed your manifest files and can provide the following recommendations:

Property	Recommendation	Issues Found
commerceSuiteVersion	SAP recommends to not specify the patch version to always have the latest security patches applied with each build. This may lead to inconsistent testing between your local development and Cloud environments when new patches are released without noticing. Customer is specifying a fixed patch version, so you must make sure you manually keep track of patches.	Yes, we highly recommend an upgrade of the platform to 2205 or to a supported version.
extensions	Solr Extension is still part of the deployment package. We highly recommend removing it from the deployment package.	Yes, Solr extension is part of the list of extensions.
properties	It is recommended that you do not include properties in your manifest file. These should be defined in the correct *.properties file and loaded from source as part of https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/1be46286b36a4aa48205be5a96240672/2311d89eef9344fc81ef168ac9668307.html?locale=en-US&version=v2105	No
aspects	For each aspect listed below the webapp was found to be defined incorrectly. Consider moving to the suggested aspect: No issues found	No

See

https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/1be46286b36a4aa48205be5a96240672/aa4777ef30f845008c64dae7218ac82d.html?locale=en-US&version=v1905

Example: How to exclude Solr Extension:

```
"useConfig": {
  "extensions": {
    "location": "env/env-development/hybris/config/localextensions.xml",
    "exclude": [
      "solrserver"
    ]
  },
}
```

3.3 Cloud Portal Configuration

A review of Cloud Portal was executed, including a review of the configurations of your endpoints. The following issues were identified:

Issue	Description
Certificate Issues	Configured but some are outdated and need to be removed. create an SAP Ticket to the support team
Maintenance Pages	Maintenance page uploaded and used.
Redirects	No redirection sets were defined. We commonly see redirects for a Commerce Cloud solution. They can be defined for an endpoint, enabling vanity & marketing URLs and site migrations. Confirm it is ok to not have any redirect sets defined. See documentation
Static Artifacts	Static artifacts configured.

See

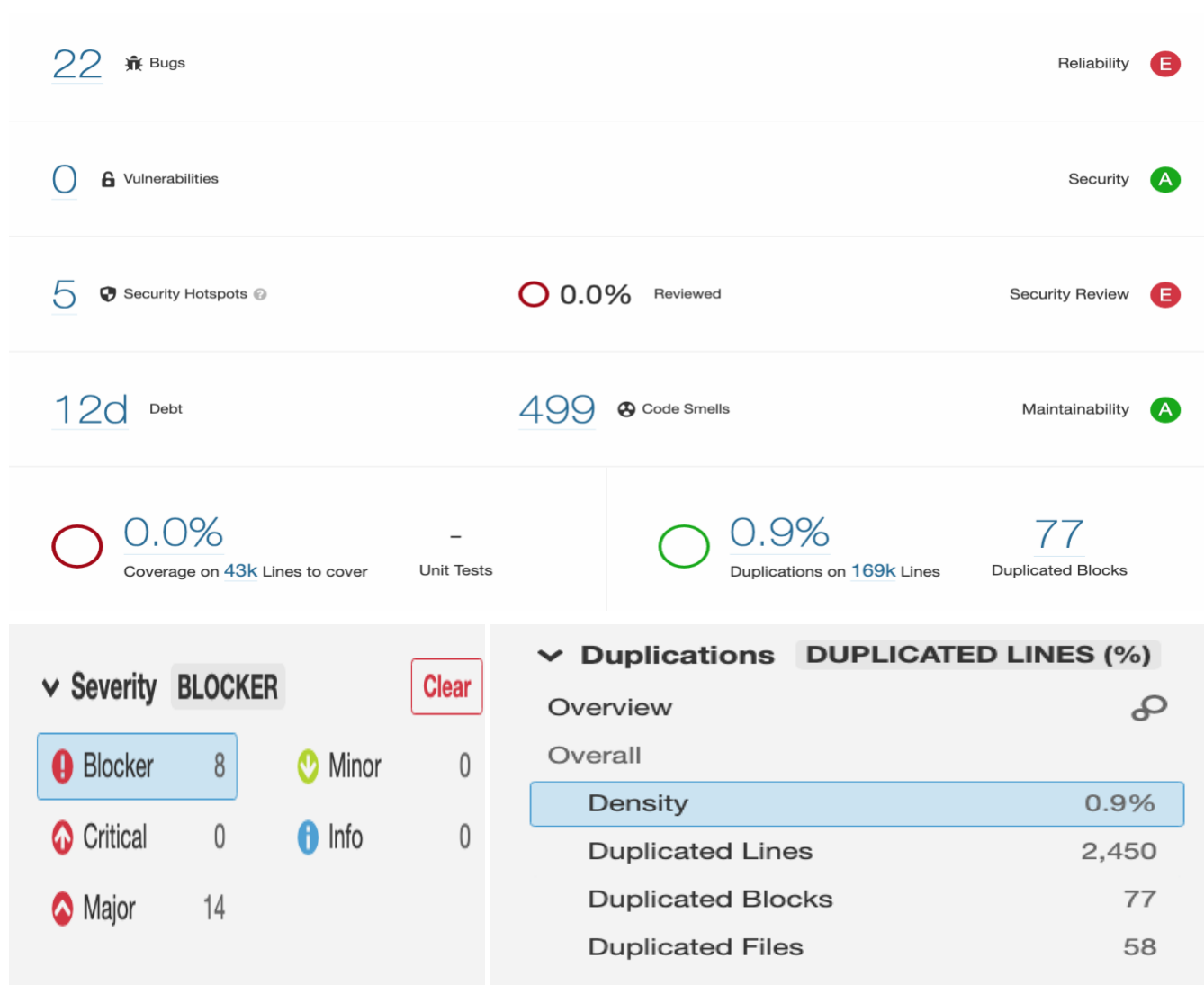
https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/0fa6bcf4736c46f78c248512391eb467/2bb13fbce5d145ec89b09ea5d2feef25.html?locale=en-US&version=v1905

for more information.

4 SOURCE CODE ANALYSIS

4.1 Quality Gates

An analysis of your custom code was executed in Sonar using the rulesets provided in the SAP Commerce Core release (e.g., java-hybris-profile.xml). We have attached the full report. For more, see https://www.sap.com/cxworks/article/2589634066/measuring_code_quality_with_sonar



Gate	Recommendation	Issues Found
Bugs & Vulnerabilities	8 blockers found. Address Blocker bugs before go-live as well as the 5 security hotspots (even though after our analysis we did not find any severe issues)	Yes
Duplication	Code duplication above 5% is not recommended. Backlog tickets should be created to deal with this	No, 3,6%
Technical Debt	Technical debt above 30 days at this point in the project could indicate underlying issues that may cause problems down the line. Backlog tickets should be created to deal with this debt as part of your development cycle. It should be monitored to ensure it's not increasing.	No, 12 days of technical debt.
Coverage	Most source code should be automatically tested. Custom classes that are not under test may run the risk of containing bugs or potentially cause increased maintenance of the project in the future. SAP recommends focusing on coverage (80%) in the Facades and Service Layers with Integration Tests. Due to problem with suspension tests, code coverage was not measured.	-

4.1.1 Bug list

Filters

Clear All Filters

Type

BUG

Clear

Bug

8

Vulnerability

0

Code Smell

0

+ click to add to selection

Severity

BLOCKER

Clear

Blocker

8

Minor

0

Critical

0

Info

0

Major

14

+ click to add to selection

Scope

Resolution

Status

Security Category

Creation Date

Language

Rule

Tag

Directory

File

Assignee

hybris/_/web/src/rs/mts/conv/SearchQueryDataConverter.java

A "NullPointerException" could be thrown; "convert()" can return null.

2 hours ago

L60

cert, cwe

hybris/_/rs/mts/request/mapping/handler/CommerceHandlerMapping.java

A "NullPointerException" could be thrown; "getApplicationContext()" can return null.

2 hours ago

L173

cert, cwe

A "NullPointerException" could be thrown; "getApplicationContext()" can return null.

2 hours ago

L177

cert, cwe

A "NullPointerException" could be thrown; "getApplicationContext()" can return null.

2 hours ago

L193

cert, cwe

hybris/_/rs/mts/facades/handler/impl/MtsMobileEntryPriceConflictStatusHandler.java

A "NullPointerException" could be thrown; "device" is nullable here.

2 hours ago

L36

cert, cwe

hybris/_/rs/mts/facades/handler/impl/MtsMobileEntryPriceForceChangedStatusHandler.java

A "NullPointerException" could be thrown; "device" is nullable here.

2 hours ago

L46

cert, cwe

hybris/_/mts/integration/service/impl/MtsCustomerAssetServiceImpl.java

A "NullPointerException" could be thrown; "importRelatedAssetForProduct()" can return null.

2 hours ago

L297

cert, cwe

A "NullPointerException" could be thrown; "getOrCreateSubscriptions()" can return null.

2 hours ago

L776

cert, cwe

8 of 8 shown

Filters

Clear All Filters

Type

BUG

Clear

Bug

14

Vulnerability

0

Code Smell

58

+ click to add to selection

Severity

MAJOR

Clear

Blocker

8

Minor

0

Critical

0

Info

0

Major

14

+ click to add to selection

Scope

Resolution

Status

Security Category

Creation Date

Language

Rule

Tag

Directory

File

Assignee

Author

hybris/_web/src/rs/mts/validator/MtsCartWithMsidnValidator.java

The return value of "orElseThrow" must be used.

2 hours ago

L46

cert

hybris/_src/rs/mts/core/order/hook/MtsTmaPoAddToCartMethodHook.java

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L158

pitfall

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L190

pitfall

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L228

pitfall

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L274

pitfall

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L291

pitfall

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L305

pitfall

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L342

pitfall

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L352

pitfall

hybris/_core/setup/datacreator/impl/C2LDataCreatorImpl.java

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L56

pitfall

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L72

pitfall

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L82

pitfall

Rename this method; there is a "private" method in the parent class with the same name.

2 hours ago

L102

pitfall

hybris/_src/rs/mts/facades/order/impl/MtsTmaCartFacadelImpl.java

The return value of "orElseThrow" must be used.

2 hours ago

L453

cert

14 of 14 shown

- Security Hotspots

5 Security Hotspots to review

Review priority: HIGH

Authentication

4

"Password" detected here, make sure this is not a hard-coded credential.

...ent/queryAvailableConfig_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.

...eDetailsByPhoneNumber_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.

...gement/sendOTASettings_Request.xml

"password" detected here, make sure this is not a hard-coded credential.

...quest/ssoUserGateway/checkUser.xml

Review priority: LOW

Others

1

"Password" detected here, make sure this is not a hard-coded credential.

Hard-coded credentials are security-sensitive [xml:S2068](#)

Status: TO REVIEW

This security hotspot needs to be reviewed to assess whether the code poses a risk.

Change status

Assignee: Not assigned

Where is the risk?

What's the risk?

Assess the risk

How can you fix it?

hybris/_src/rs/mts/score/resources/data/device-management/queryAvailableConfig_Request.xml

Open in IDE

Get Permalink

1

<soapenv:Envelope xmlns: xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns: xsd="http://www.w3.org/2001/XMLSchema" xmlns: soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns: mfs="http://services.mformation.com/MFServices/">

2

<soapenv:Header>

3

<oas:Security soapenv: mustUnderstand="1" soapenv: actor="mf" xmlns: oas="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd">

4

<oas:UsernameToken>

5

<oas:Username>\$username</oas:Username>

6

<oas:Password>\$password</oas:Password>

"Password" detected here, make sure this is not a hard-coded credential.

Comment

7

</oas:UsernameToken>

8

</oas:Security>

9

</soapenv:Header>

13

5 Security Hotspots to review

Review priority: **HIGH**

Authentication4

"Password" detected here, make sure this is not a hard-coded credential.
...ent/queryAvailableConfig_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.
...eDetailsByPhoneNumber_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.
...gement/sendOTASettings_Request.xml

"password" detected here, make sure this is not a hard-coded credential.
...quest/ssoUserGateway/checkUser.xml

Review priority: **LOW**

Others1

"Password" detected here, make sure this is not a hard-coded credential.

Hard-coded credentials are security-sensitive xml:S2068

Status: **TO REVIEW**

This security hotspot needs to be reviewed to assess whether the code poses a risk.

Change status

Assignee: Not assigned

Where is the risk?	What's the risk?	Assess the risk	How can you fix it?
hybris/.../mts/mtscore/resources/data/device-management/queryDeviceDetailsByPhoneNumber_Request.xml Open in IDEGet Permalink 1<soapenv:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:mfs="http://services.mformation.com/MFServices/"> 2<soapenv:Header> 3<oas:Security soapenv:mustUnderstand="1" soapenv:actor="mf" xmlns:oas="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd"> 4<oas:UsernameToken> 5<oas:Username>Username</oas:Username> 6<oas:Password>password</oas:Password> "Password" detected here, make sure this is not a hard-coded credential. Comment 7</oas:UsernameToken> 8</oas:Security> 9</soapenv:Header>			

5 Security Hotspots to review

Review priority: **HIGH**

Authentication4

"Password" detected here, make sure this is not a hard-coded credential.
...ent/queryAvailableConfig_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.
...eDetailsByPhoneNumber_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.
...gement/sendOTASettings_Request.xml

"password" detected here, make sure this is not a hard-coded credential.
...quest/ssoUserGateway/checkUser.xml

Review priority: **LOW**

Others1

"Password" detected here, make sure this is not a hard-coded credential.

Hard-coded credentials are security-sensitive xml:S2068

Status: **TO REVIEW**

This security hotspot needs to be reviewed to assess whether the code poses a risk.

Change status

Assignee: Not assigned

Where is the risk?	What's the risk?	Assess the risk	How can you fix it?
hybris/.../mts/mtscore/resources/data/device-management/sendOTASettings_Request.xml Open in IDEGet Permalink 1<soapenv:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:mfs="http://services.mformation.com/MFServices/"> 2<soapenv:Header> 3<oas:Security soapenv:mustUnderstand="1" soapenv:actor="mf" xmlns:oas="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd"> 4<oas:UsernameToken> 5<oas:Username>Username</oas:Username> 6<oas:Password>password</oas:Password> "Password" detected here, make sure this is not a hard-coded credential. Comment 7</oas:UsernameToken> 8</oas:Security> 9</soapenv:Header>			

5 Security Hotspots to review

Review priority: **HIGH**

Authentication4

"Password" detected here, make sure this is not a hard-coded credential.
...ent/queryAvailableConfig_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.
...eDetailsByPhoneNumber_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.
...gement/sendOTASettings_Request.xml

"password" detected here, make sure this is not a hard-coded credential.
...quest/ssoUserGateway/checkUser.xml

Review priority: **LOW**

Others1

"password" detected here, make sure this is not a hard-coded credential.

Hard-coded credentials are security-sensitive xml:S2068

Status: **TO REVIEW**

This security hotspot needs to be reviewed to assess whether the code poses a risk.

Change status

Assignee: Not assigned

Where is the risk?	What's the risk?	Assess the risk	How can you fix it?
hybris/.../resources/template/request/ssoUserGateway/checkUser.xml Open in IDEGet Permalink 1<ns:CheckUser xmlns:ns="http://tempuri.org/"> 2<ns:username>some_value</ns:username> 3<!--Optional:--> 4<ns:password>some_value</ns:password> "password" detected here, make sure this is not a hard-coded credential. Comment 5</ns:CheckUser>			

5 Security Hotspots to review

Review priority: **HIGH**

Authentication 4

"Password" detected here, make sure this is not a hard-coded credential.

...ent/queryAvailableConfig_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.

...eDetailsByPhoneNumber_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.

...gement/sendOTASettings_Request.xml

"password" detected here, make sure this is not a hard-coded credential.

...quest/ssoUserGateway/checkUser.xml

"password" detected here, make sure this is not a hard-coded credential.

Hard-coded credentials are security-sensitive `xml:S2068`

Status: **TO REVIEW**

This security hotspot needs to be reviewed to assess whether the code poses a risk.

Change status

Assignee: Not assigned

Where is the risk?

What's the risk?

Assess the risk

How can you fix it?

hybris/...resources/template/request/ssoUserGateway/checkUser.xml

Open in IDE

Get Permalink

```

1 <ns:CheckUser xmlns:ns="http://tempuri.org/">
2   <ns:username>some_value</ns:username>
3   <!--Optional!-->
4   <ns:password>some_value</ns:password>
5 </ns:CheckUser>

```

"password" detected here, make sure this is not a hard-coded credential.

Comment

5 Security Hotspots to review

Review priority: **HIGH**

Authentication 4

"Password" detected here, make sure this is not a hard-coded credential.

...ent/queryAvailableConfig_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.

...eDetailsByPhoneNumber_Request.xml

"Password" detected here, make sure this is not a hard-coded credential.

...gement/sendOTASettings_Request.xml

"password" detected here, make sure this is not a hard-coded credential.

...quest/ssoUserGateway/checkUser.xml

Review priority: **LOW**

Others 1

Make sure using this hardcoded IP address is safe here.

...etCallHistoryPerAssetHealthCheck.java

Make sure using this hardcoded IP address is safe here.

Using hardcoded IP addresses is security-sensitive `java:S1313`

Status: **TO REVIEW**

This security hotspot needs to be reviewed to assess whether the code poses a risk.

Change status

Assignee: Not assigned

Where is the risk?

What's the risk?

Assess the risk

How can you fix it?

hybris/...src/rs/integration/healthcheck/MtsGetCallHistoryPerAssetHealthCheck.java

Open in IDE

Get Permalink

```

34 * @author Katsiaryna Karashkova <k.karashkova@sam-solutions.com>
35 */
36 public class MtsGetCallHistoryPerAssetHealthCheck extends AbstractMtsHealthCheck
37 {
38     private static final String CHECK_NAME = "Get call history per asset";
39     private static final int MONTH = 0;
40     private static final String SERVICE_TYPE = "CALLS";
41     private static final String IP_ADDRESS = "10.118.76.221";
42
43     @Resource(name = "invoiceManagementWSGateway")
44     private MtsInvoiceManagementClient invoiceManagementClient;
45

```

Make sure using this hardcoded IP address is safe here.

Comment

Comment:

Hard-coded IP Addresses are generally not a good idea because doing so, we introduce Infrastructure concerns into application concerns. If the IP were to change, the health check would no longer work. Please review the piece of code (see below):

4.1.2 Reliability

MTS ☆ master October 18, 2022 at 12:51 AM Version MTS

Overview Issues Security Hotspots Measures Code Activity Project Settings Project Information

Project Overview

MTS / hybris/bin/custom/mts View as Tree ↑ ↓ to select files ← → to navigate 12 files

Reliability RELIABILITY REMED...
Overview
Overall
Bugs 22
Rating E
Remediation Effort 7h 40min

Security
Security Review
Maintainability
Coverage
Duplications
Overview
Overall
Density 0.9%
Duplicated Lines 2,450
Duplicated Blocks 77
Duplicated Files 58

Reliability Remediation Effort 7h 40min

log4jpatch	0
mtsbackoffice	0
mtsscommercewebservices	50min
mtssconsentintegration	0
mtsscore	6h
mtssfacades	30min
mtsshac	0
mtsinitialdata	0
mtsintegration	20min
mtssoracledbintegration	0
mtsspatches	0
mtstest	0

12 of 12 shown

4.2 OWASP Security Standards

Sonar has identified no potential security issues.

▼ **Security Category**

▼ **SonarSource**

▼ **OWASP Top 10 2021**
No results

▼ **OWASP Top 10 2017**
No results

▼ **SANS Top 25**
No results

► **CWE**

4.3 Type System Validation

The Type System Validation (TSV) Tool is a tool created by SAP Services to evaluate custom types in your solution. The following items in the type system must be investigated:

FILENAME	LINE	RULE	PRIORITY	ELEMENT	DESCRIPTION
mtscore-items.xml	3784	MandatoryFieldMustHaveInitialValue	M	MtsMWallet.msisdn	Mandatory fields (where optional='false') must either have initial set to 'true' or a default value defined
mtscore-items.xml	3807	BooleanFieldCannotBeOptional	M	MtsMWallet.accountLinked	Fields of type java.lang.Boolean must be mandatory (optional='false') or a default value defined
mtscore-items.xml	504	OrderingOfRelationShouldBeAvoided	L	MtsTmaProductSpecCharacteristicGroup2TmaProductSpecCharacteristicRelation.productSpecCharacteristics	Any side of a relation that has cardinality='many' should not have ordered='true' unless absolutely necessary
mtscore-items.xml	613	OrderingOfRelationShouldBeAvoided	L	MtsSolrSearchQueryTemplate2SolrSortRel.sorts	Any side of a relation that has cardinality='many' should not have ordered='true' unless absolutely necessary
mtscore-items.xml	504	ListsInRelationShouldBeAvoided	L	MtsTmaProductSpecCharacteristicGroup2TmaProductSpecCharacteristicRelation.productSpecCharacteristics	Any side of a relation that has cardinality='many' should have collectiontype='set' unless absolutely necessary
mtscore-items.xml	535	ListsInRelationShouldBeAvoided	L	DocumentMWalletEmailProcess2EmailAttachmentsRel.attachments	Any side of a relation that has cardinality='many' should have collectiontype='set' unless absolutely necessary
mtscore-items.xml	613	ListsInRelationShouldBeAvoided	L	MtsSolrSearchQueryTemplate2SolrSortRel.sorts	Any side of a relation that has cardinality='many' should have collectiontype='set' unless absolutely necessary
mtscore-items.xml	644	PersistenceColumnMustBeForAllSupportedDatabases	M	number	If a columntype tag is specified for an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,hsql,db,sap).
mtscore-items.xml	654	PersistenceColumnMustBeForAllSupportedDatabases	M	longDescription	If a columntype tag is specified for an attribute's persistence, it has to be defined for all supported

		IISupport edDatab ases			databases(oracle,mysql,sqlserver,h sqldb,sap).
mtscore- items.xml	818	Persisten ceColum nTypeMu stBeForA IISupport edDatab ases	M	decryptablePa ssword	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqldb,sap).
mtscore- items.xml	896	Persisten ceColum nTypeMu stBeForA IISupport edDatab ases	M	descriptionLin e	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqldb,sap).
mtscore- items.xml	1622	Persisten ceColum nTypeMu stBeForA IISupport edDatab ases	M	balkanMusicInf oMessage	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqldb,sap).
mtscore- items.xml	1700	Persisten ceColum nTypeMu stBeForA IISupport edDatab ases	M	eSIMDescripti onLine	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqldb,sap).
mtscore- items.xml	1835	Persisten ceColum nTypeMu stBeForA IISupport edDatab ases	M	oneInBoxThre eLongDescripti on	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqldb,sap).
mtscore- items.xml	1869	Persisten ceColum nTypeMu stBeForA IISupport edDatab ases	M	fourInBoxFour LongDescriptio n	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqldb,sap).
mtscore- items.xml	2094	Persisten ceColum nTypeMu stBeForA IISupport edDatab ases	M	benefits	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqldb,sap).
mtscore- items.xml	2264	Persisten ceColum nTypeMu stBeForA IISupport	M	commerceDes criptionLine	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqldb,sap).

		edDatab ases			
mtscore- items.xml	2441	Persisten ceColum nTypeMu stBeForA llSupport edDatab ases	M	additionalInfoL ine	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqlldb,sap).
mtscore- items.xml	2449	Persisten ceColum nTypeMu stBeForA llSupport edDatab ases	M	additionalInfoT ooltip	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqlldb,sap).
mtscore- items.xml	2507	Persisten ceColum nTypeMu stBeForA llSupport edDatab ases	M	description	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqlldb,sap).
mtscore- items.xml	2793	Persisten ceColum nTypeMu stBeForA llSupport edDatab ases	M	descriptionLin e	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqlldb,sap).
mtscore- items.xml	3881	Persisten ceColum nTypeMu stBeForA llSupport edDatab ases	M	additionalInfoL ine	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqlldb,sap).
mtscore- items.xml	3889	Persisten ceColum nTypeMu stBeForA llSupport edDatab ases	M	descriptionExt	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqlldb,sap).
mtscore- items.xml	3933	Persisten ceColum nTypeMu stBeForA llSupport edDatab ases	M	description	If a columntype tag is specified for a an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,h sqlldb,sap).
mtscore- items.xml	1694	Collectio nsAreOnl yForDyn amicAnd Jalo	L	kosovoCodes	Collections have to be only defined for dynamic and jalo attributes

mtscore-items.xml	3684	TypeHasAtLeastOneIndexSpecified	M	MtsSearchQueryTemplateSort	Not at least one database indexes is defined for type.
mtscore-items.xml	3756	TypeHasAtLeastOneIndexSpecified	M	MtsTisNetSpeed	Not at least one database indexes is defined for type.
mtscore-items.xml	3992	TypeHasAtLeastOneIndexSpecified	M	MtsChannelArchiveHistory	Not at least one database indexes is defined for type.
mtscore-items.xml	701	NoDatabaseIndexesForUniqueAttributesOfType	M	Customer	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	2188	NoDatabaseIndexesForUniqueAttributesOfType	M	TmaSimpleProductOffering	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	2297	NoDatabaseIndexesForUniqueAttributesOfType	M	TmaPoVariant	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	2573	NoDatabaseIndexesForUniqueAttributesOfType	M	ProductReference	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	2749	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsGiftProductReference	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3096	NoDatabaseIndexesForUniqueAttributesOfType	M	PaginatedCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3106	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsCleanUpNotRegisteredUsersCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.

mtscore-items.xml	3122	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsProcessLockTransactionCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3134	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsProcessCheckPaymentBillsCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3147	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsProcessCheckPaymentToPUpCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3159	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsProcessCheckPaymentMtsTVGoAddonCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3172	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsResendPaymentCaptureCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3184	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsResendToPUpCaptureCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3196	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsResendMtsTVGoAddonCaptureCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3208	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsDeactivateMtsTVGoAddonAfterPaymentFailureCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3222	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsCleanUpCustomerDecryptablePasswordCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.

mtscore-items.xml	3233	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsInstallationAddressResetCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3236	NoDatabaseIndexesForUniqueAttributesOfType	M	CronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3243	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsCleanUpInvalidOtpAccessTokensCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3352	NoDatabaseIndexesForUniqueAttributesOfType	M	AbstractOrderEntry	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3397	NoDatabaseIndexesForUniqueAttributesOfType	M	Order	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3418	NoDatabaseIndexesForUniqueAttributesOfType	M	AbstractOrder	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3684	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsSearchQueryTemplateSort	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	3776	NoDatabaseIndexesForUniqueAttributesOfType	M	MtsMWallet	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscore-items.xml	550	One2ManyRelationWhereTheOneSideIsCatalogAwareTypeAndTheMany	H	MtsGiftProductReference2SPRelation	One 2 many relation where the one side is a Catalog aware type and the many side is not

		SidelsNot			
mtscore-items.xml	1133	NoUniqueAttributesDefinedForType	M	MtsBillDetails	No unique attributes defined for type
mtscore-items.xml	3258	NoUniqueAttributesDefinedForType	M	MtsOtpRequest	No unique attributes defined for type
mtscore-items.xml	3317	NoUniqueAttributesDefinedForType	M	MtsTopUpCheck	No unique attributes defined for type
mtscore-items.xml	3756	NoUniqueAttributesDefinedForType	M	MtsTisNetSpeed	No unique attributes defined for type
mtscore-items.xml	3910	NoUniqueAttributesDefinedForType	M	MtsOrderConsent	No unique attributes defined for type
mtscore-items.xml	3922	NoUniqueAttributesDefinedForType	M	MtsChannelProgram	No unique attributes defined for type
mtscore-items.xml	3992	NoUniqueAttributesDefinedForType	M	MtsChannelArchiveHistory	No unique attributes defined for type
mtscore-items.xml	2987	UniqueIndexUsingNonUniqueAttributes	M	PaymentTransactionEntry	A unique index should contain only unique attributes cause it limits duplicate inserts to be not possible in that combination.
mtscore-items.xml	3007	UniqueIndexUsingNonUniqueAttributes	M	PaymentTransaction	A unique index should contain only unique attributes cause it limits duplicate inserts to be not possible in that combination.
mtscore-items.xml	3580	UniqueIndexUsingNonUniqueAttributes	M	OAuthAccessToken	A unique index should contain only unique attributes cause it limits duplicate inserts to be not possible in that combination.
mtscore-items.xml	3877	UniqueIndexUsingNonUniqueAttributes	M	ConsentTemplate	A unique index should contain only unique attributes cause it limits duplicate inserts to be not possible in that combination.
mtscore-items.xml	3910	UniqueIndexUsingNonUniqueAttributes	M	MtsOrderConsent	A unique index should contain only unique attributes cause it limits duplicate inserts to be not possible in that combination.

mtsintegration-items.xml	43	PersistenceColumnTypesMustBeForAllSupportedDatabases	M	message	If a columntype tag is specified for an attribute's persistence, it has to be defined for all supported databases(oracle,mysql,sqlserver,hsqldb,sap).
mtsintegration-items.xml	124	CollectionsAreOnlyForDynamicAndJalo	L	products	Collections have to be only defined for dynamic and jalo attributes
mtsintegration-items.xml	32	TypeHasAtLeastOneIndexSpecified	M	DatabaseLogEntry	Not at least one database indexes is defined for type.
mtsintegration-items.xml	111	NoDatabaseIndexesForUniqueAttributesOfType	M	UpdateCustomerInfoCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtsintegration-items.xml	122	NoDatabaseIndexesForUniqueAttributesOfType	M	UpdateRecommendationCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtsintegration-items.xml	32	NoUniqueAttributesDefinedForType	M	DatabaseLogEntry	No unique attributes defined for type
mtscommerceservices-items.xml	44	CollectionsAreOnlyForDynamicAndJalo	L	sites	Collections have to be only defined for dynamic and jalo attributes
mtscommerceservices-items.xml	19	NoDatabaseIndexesForUniqueAttributesOfType	M	ProductExpressUpdateCleanerCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscommerceservices-items.xml	30	NoDatabaseIndexesForUniqueAttributesOfType	M	OrderStatusUpdateCleanerCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.
mtscommerceservices-items.xml	42	NoDatabaseIndexesForUniqueAttributesOfType	M	OldCartRemovalCronJob	No database indexes defined for unique attributes of type. Mostly it makes sense to have it in place as it is used as main WHERE clause in queries and there can speedup fetching entries.

4.4 Database Compatibility

Some attributes in your data model have been declared with database specific mapping that may not be correct for Microsoft Azure SQL database used by Commerce Cloud. In the help page https://help.sap.com/docs/SAP_COMMERCE/d0224eca81e249cb821f2cdf45a82ace/8c6254f086691014b095a08a61d1efed.html?locale=en-US&version=2105

under section 'Advanced Deployment' you will find further information's and the standard mapping of HYBRIS.LONG_STRING to the dedicated RDBMS data types and the hint to core-advanced-deployment.xml in the 'Advanced Deployment' section.

Our analysis did not find any issue.

Data Type	Database Mapping			
	sap	hsqldb	mysql	sqlserver
java.lang.String	nvarchar(255)	VARCHAR(255)	varchar(255)	nvarchar(255)
String	nvarchar(255)	VARCHAR(255)	varchar(255)	nvarchar(255)
java.lang.Float	decimal(20,5)	float	float(20,5)	float
java.lang.Double	decimal(30,8)	double	double	float
java.lang.Byte	smallint	smallint	smallint	integer
java.lang.Character	smallint	smallint	smallint	char(4)
java.lang.Short	integer	smallint	integer	integer
java.lang.Boolean	decimal(1,0)	tinyint	tinyint(1)	tinyint
java.lang.Long	bigint	bigint	bigint	bigint
java.lang.Integer	bigint	int	integer	integer
float	decimal(20,5) DEFAULT 0	float default 0	float(20,5) DEFAULT 0	float default 0
double	decimal(30,8) DEFAULT 0	double default 0	double DEFAULT 0	float default 0
byte	integer DEFAULT 0	smallint default 0	smallint DEFAULT 0	integer default 0
char	integer DEFAULT "	smallint default 0	smallint DEFAULT 0	char(4) default "
short	integer DEFAULT 0	smallint default 0	integer DEFAULT 0	integer default 0
boolean	decimal(1,0) DEFAULT 0	tinyint default 0	tinyint(1) DEFAULT 0	tinyint default 0
long	bigint DEFAULT 0	bigint default 0	bigint DEFAULT 0	integer default 0
int	bigint DEFAULT 0	int default 0	integer DEFAULT 0	integer default 0
java.util.Date	timestamp	timestamp	datetime	datetime2
java.math.BigDecimal	decimal(30,8)	decimal(30,8)	decimal(30,8)	decimal(30,8)
java.io.Serializable	blob	longvarbinary	longblob	image
HYBRIS.LONG_STRING	nvarchar(5000)	longvarchar	text	nvarchar(max)
HYBRIS.JSON	nclob	longvarchar	longtext	nvarchar(max)
HYBRIS.COMMA_SEPARATED_PKS	nvarchar(5000)	longvarchar	text	nvarchar(max)
HYBRIS.PK	bigint	BIGINT	bigint	bigint

4.5 Search Restriction

All active search restrictions were reviewed. The following search restrictions should be reviewed to ensure they follow recommended practices

Identifier	Active	Restricted Type	Apply on
HideSystemPrincipals	TRUE	Principal	[employeeegroup]
Frontend_Category	TRUE	Category	[customergroup]
Frontend_ProductOfflineDate	TRUE	Product	[customergroup]
Frontend_ProductApprovalStatus	TRUE	Product	[customergroup]
Frontend_Product	TRUE	Product	[customergroup]
Frontend_Keyword	TRUE	Keyword	[customergroup]
Frontend_ProductReference	TRUE	ProductReference	[customergroup]
Backend_visibility	TRUE	TmaServiceSpecification	[employeeegroup]
Backend_visibility	TRUE	MtsBuyingRuleGroup	[employeeegroup]
Backend_visibility	TRUE	TmaCompatibilityPolicy	[employeeegroup]
Backend_visibility	TRUE	CMSRelation	[employeeegroup]
Backend_visibility	TRUE	ClassificationAttributeUnit	[employeeegroup]
Backend_visibility	TRUE	DiscountRow	[employeeegroup]
Backend_visibility	TRUE	TmaPolicyCondition	[employeeegroup]
Backend_visibility	TRUE	AbstractConfiguratorSetting	[employeeegroup]
Backend_visibility	TRUE	UsageCharge	[employeeegroup]
Backend_visibility	TRUE	TmaProductSpecCharacteristic	[employeeegroup]
Backend_visibility	TRUE	TaxRow	[employeeegroup]
Backend_visibility	TRUE	TmaProductSpecification	[employeeegroup]
Backend_visibility	TRUE	ClassificationAttributeValue	[employeeegroup]
Backend_visibility	TRUE	ChargeEntry	[employeeegroup]
Backend_visibility	TRUE	MtsTmaProductSpecCharacteristicGroup	[employeeegroup]

Backend_visibility	TRUE	AbstractAsSearchConfiguration	[employeeegroup]
Backend_visibility	TRUE	TmaPolicyAction	[employeeegroup]
Backend_visibility	TRUE	TmaBpoPreConfig	[employeeegroup]
Backend_visibility	TRUE	PriceRow	[employeeegroup]
Backend_visibility	TRUE	ClassificationAttribute	[employeeegroup]
Backend_visibility	TRUE	TmaProductSpecCharacteristicValue	[employeeegroup]
Backend_visibility	TRUE	TmaServiceSpecCharacteristicValue	[employeeegroup]
Backend_visibility	TRUE	AttributeValueAssignment	[employeeegroup]
Backend_visibility	TRUE	CMSItem	[employeeegroup]
Backend_visibility	TRUE	TmaPolicyStatement	[employeeegroup]
Backend_visibility	TRUE	Media	[employeeegroup]
Backend_visibility	TRUE	MtsDeviceInsurance	[employeeegroup]
Backend_visibility	TRUE	TmaServiceSpecCharacteristic	[employeeegroup]
Backend_visibility	TRUE	AsSearchProfileActivationSet	[employeeegroup]
Backend_visibility	TRUE	Category	[employeeegroup]
Backend_visibility	TRUE	ClassAttributeAssignment	[employeeegroup]
Backend_visibility	TRUE	Keyword	[employeeegroup]
Backend_visibility	TRUE	AbstractAsSearchProfile	[employeeegroup]
Backend_visibility	TRUE	AbstractAsItemConfiguration	[employeeegroup]
Backend_visibility	TRUE	MediaContainer	[employeeegroup]
Backend_visibility	TRUE	TmaProductOfferingGroup	[employeeegroup]
Backend_visibility	TRUE	Product	[employeeegroup]
Backend_visibility	TRUE	VariantProduct	[employeeegroup]
Backend_visibility	TRUE	Catalog	[employeeegroup]

Backend_visibility	TRUE	CatalogVersion	[employeeegroup]
Frontend_Group_PriceRows	TRUE	PriceRow	[customergroup]
Frontend_Group_TaxRows	TRUE	TaxRow	[customergroup]
Frontend_Group_DiscountRows	TRUE	DiscountRow	[customergroup]
Backoffice_Media_Catalog	TRUE	Catalog	[employeeegroup]
Backoffice_Configuration_Media	TRUE	Media	[employeeegroup]
Backoffice_Media_Catalog_Version	TRUE	CatalogVersion	[employeeegroup]
CustomerReviewRestrictions	TRUE	CustomerReview	[customergroup]
outboundMonitoringIntegrationVisi bility	TRUE	IntegrationObject	Integration User Group [integrationusergroup]
integrationServiceVisibility	TRUE	IntegrationObject	Integration User Group [integrationusergroup]
inboundMonitoringIntegrationVisibi lity	TRUE	IntegrationObject	Integration User Group [integrationusergroup]
promotion_rule_type_visibility	TRUE	RuleType	[marketingManagerGroup]
promotion_drools_kie_module_visi bility	TRUE	DroolsKIEModule	[marketingManagerGroup]
insuranceCatalogVersionRestrictio n	TRUE	MtsDeviceInsurance	[customergroup]
mobileSubscriptionWithoutPaymen tStatusRestriction	TRUE	TmaSubscriptionBase	[customergroup]
sppOnlineRestriction	TRUE	SubscriptionPricePlan	[customergroup]
MediaContainerProductContentMan agerRestriction	TRUE	MediaContainer	Product Content Manager group [productcontentmanagergroup]
mediaProductContentManagerRestr iction	TRUE	Media	Product Content Manager group [productcontentmanagergroup]
specCharacteristicValueProductCo ntentManagerRestriction	TRUE	TmaProductSpecCharact eristicValue	Product Content Manager group [productcontentmanagergroup]
TmaProductSpecificationProductCo ntentManagerRestriction	TRUE	TmaProductSpecification	Product Content Manager group [productcontentmanagergroup]
catalogProductContentManagerRes triction	TRUE	Category	Product Content Manager group [productcontentmanagergroup]
productModelProductContentMana gerRestriction	TRUE	Product	Product Content Manager group [productcontentmanagergroup]
FrontEnd_Reviews	TRUE	CustomerReview	[customergroup]

4.6 Critical Business Processes

https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/aa417173fe4a4ba5a473c93eb730a417/8b9ce4868669101499b2f0f25ef9395f.html?locale=en-US&version=v1905

The cronjobs are commonly used for triggering repeated tasks at a scheduled time.

A key consideration when preparing to use the SAP Commerce Cloud is to ensure that any bulk processing jobs operate reliably within the fixed resource footprint of the 'backgroundProcessing'.

For CCv2 backgroundProcessing node has by default defined property value:

```
cluster.node.groups=backgroundProcessing,integration,yHotfolderCandidate
```

Having that all cron jobs in system should have set nodeGroup attribute to "backgroundProcessing". It is possible to check project data and check ImpEx files if they have nodeGroup set. e.g.

```
INSERT_UPDATE CronJob; code[unique=true];job(code);nodeGroup  
;myCronJob;myJob;backgroundProcessing
```

or by running FlexibleSearch in hac

```
-- code responsible for listing all cron jobs in database which is not assigned  
to backgroundProcessing node group  
select * from {CronJob} where {nodeGroup} is not null
```

That validation is required due to CCv2 task engine configuration, which is enabled for 2 aspects: backgroundProcessing and backoffice. It is possible to check by looking up for configuration in hac:

```
task.engine.loadonstartup=true
```

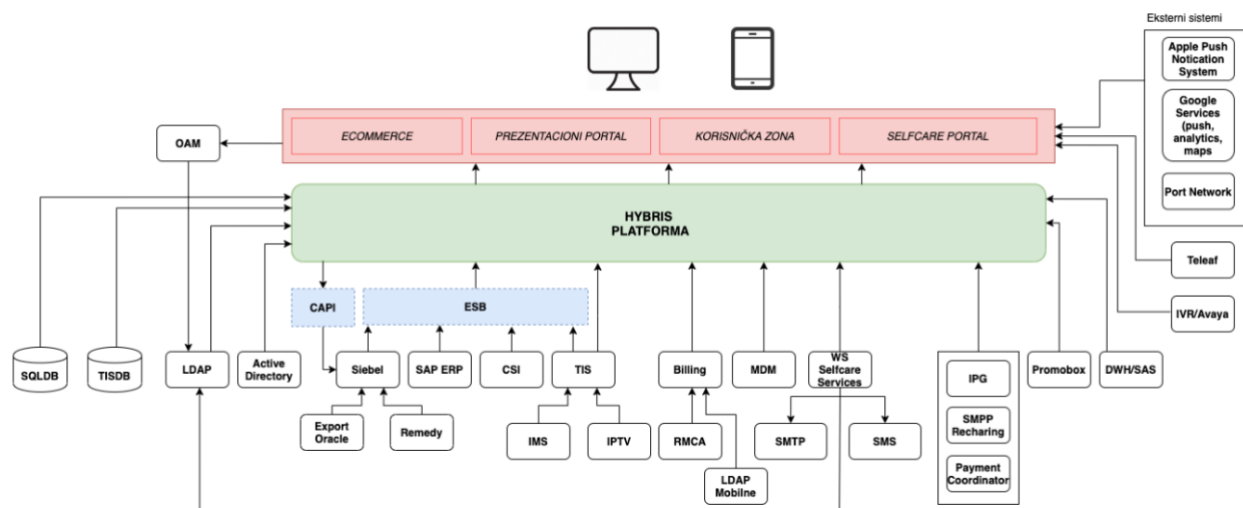
All possible cron jobs should be run on backgroundProcessing only, and not interfere with backoffice activities. Due to backoffice specific implementation it is required to have task engine enabled to be able to run backoffice's specific job: update-backofficeIndex-CronJob

(See attachment for the details)

5 INTEGRATIONS

5.1 High Level Architecture

Below is a high-level architecture diagram used only for reference:



5.2 Integration Practices

We have completed an overall assessment and have outlined the following considerations:

A lot of integration points are part of the landscape. However, if the different systems are part of the on-premises landscape, we highly recommend considering introducing VPN to secure connections to the hosted systems.

For more information,

https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/0fa6bcf4736c46f78c248512391eb467/f63dfaedd22d949ed9aadbb7835584586.html?locale=en-US&version=v1905

Integration points are necessary evil which means while they are very necessary, their issues can quickly propagate to SAP Commerce Cloud and potentially lead to application crashes or performance degradation. Every integration point will eventually fail in some way, and you need to be prepared for that failure. Integration point failures take several forms, ranging from various network errors to semantic errors. You will not get nice error responses delivered through the defined protocol; instead, you'll see protocol violation, slow response, or outright hang.

Failure in a remote system quickly becomes your problem, usually as a cascading failure when your code isn't defensive enough.

We recommend considering protecting integration points with the following patterns:

- The use of timeouts: The Timeouts pattern prevents calls to Integration Points from becoming Blocked Threads.
- Circuit breaker: protects systems from all manner of Integration Points problems. When there's a difficulty with Integration Points, calls should be stopped.
- Bulkheads: The Bulkheads pattern partitions capacity to preserve partial functionality when bad things happen.

Consider using the following stability patterns to reinforce stability, timeouts, circuit breaker and bulkheads. There are several open source projects such as Hystrix(<https://github.com/Netflix/Hystrix>) or Resiliency4J(<https://github.com/resilience4j/resilience4j>) .

With SAP Commerce Cloud,

https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/0fa6bcf4736c46f78c248512391eb467/249b3a2b0a2a469e8e18c8bbc6b77206.html?locale=en-US&version=v1905

comes embedded as part of every subscription, giving you access to a powerful set of tools to analyze and optimize the performance of your site.

Before proceeding it is recommended to review the features and capabilities in Dynatrace by consulting the

https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/0fa6bcf4736c46f78c248512391eb467/249b3a2b0a2a469e8e18c8bbc6b77206.html?locale=en-US&version=v1905

section of the product documentation. You should ensure you're comfortable with:

1. Setting up alerts
2. If you require alerts via email, please contact SAP Support.
3. Understanding service flows
4. Viewing existing dashboards and creating your own

For information:

https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/6065ea129a544adba2c3e0791dc069f2/070bb2ee3fe9479abfe2a1a107c34a13.html?locale=en-US&version=v1905

https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/6065ea129a544adba2c3e0791dc069f2/6117a684a36c4689b759bf2b5a8d0a61.html?locale=en-US&version=v1905

https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/6065ea129a544adba2c3e0791dc069f2/0c77e2eb0bf6484894906b7ebaaa25c8.html?locale=en-US&version=v1905

6 TEST APPROACH

6.1 Test Data

It is recommended to have realistic test data for load testing your application.

6.2 Performance Testing

Performance testing is the most accurate way to predict how your production environment will handle your initial go-live and subsequent weeks.

Executing performance tests is a very important step to safeguard the go-live. Please perform this step before go-live and compare results with previous environment setup to make sure, that CCv2 setup won't make things worse.

Running performance tests is out of the scope of the Cloud Readiness Check, but we recommend you review https://www.sap.com/cxworks/article/458163553/managing_performance_in_an_sap_commerce_cloud_project and become familiar with

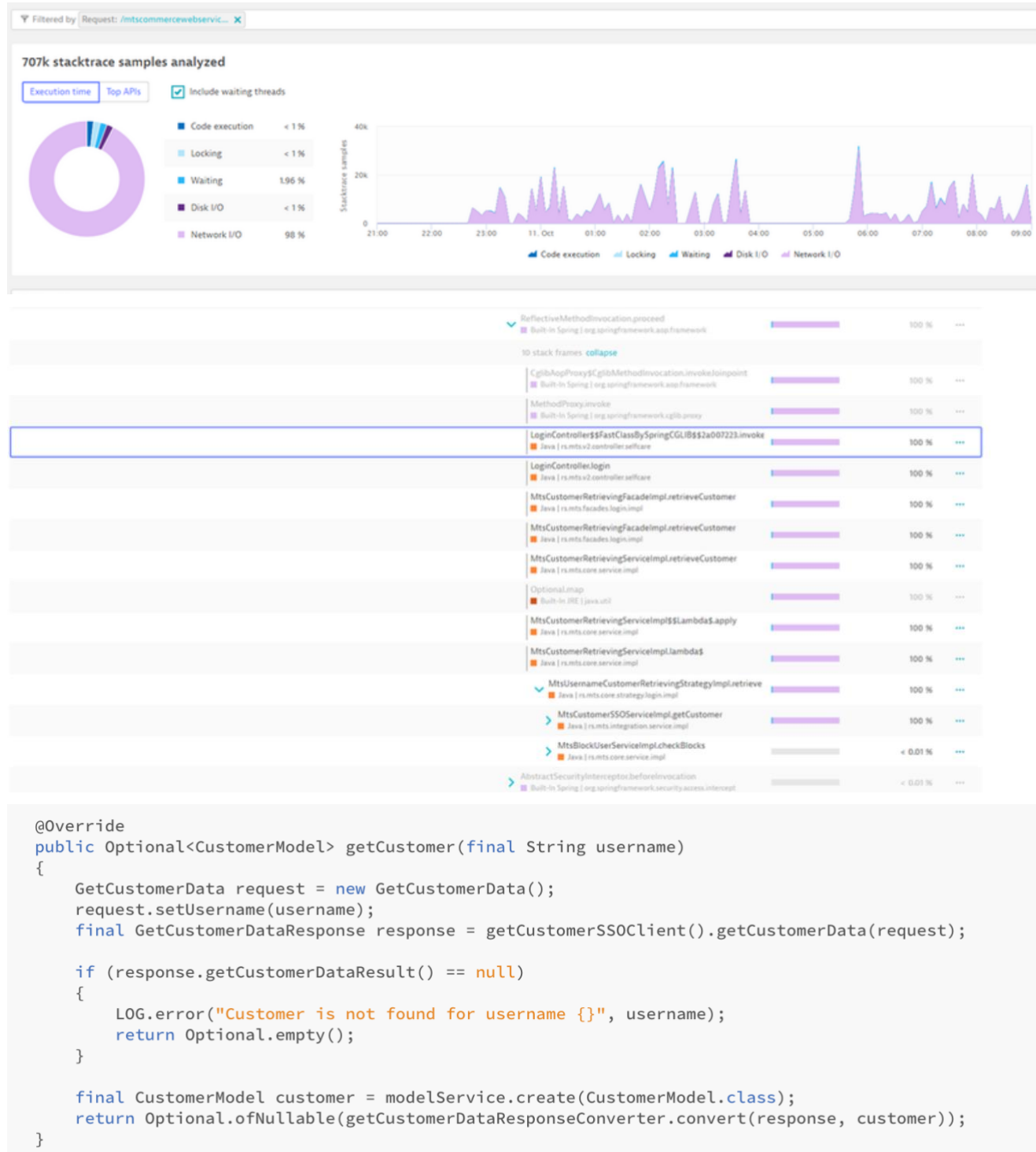
https://www.sap.com/cxworks/article/2589632460/using_dynatrace_to_manage_and_optimize_the_performance_of_your_sap_commerce_cloud_solution

The project has provided some results of performance tests. We've made the following observations:

- In the results, it is mentioned that there were some issues with the VPN.
 - o Is the VPN part of the production environment?
 - o It is recommended to perform tests ideally in an environment which is similar to the production.
- The slowest requests are login and customer search.

Name	Total time consumption	Slowest 5%	Median response time	Actions
/mtscommerceservices/v2/mtsb2cSelfcare/auth/login	1.06 min	1 min		
Customer Search - BOSearchTa...	56.9 s	2.7 s		
/mtscommerceservices/v2/mtsb2c/users/current	3.26 s	1.06 s		
SubscriptionPricePlan Search OrderValue PlaceOrder AddToCart OrderItemC... Currency IsGuestChec... BOSearchTa... NumberOfS...	1.31 s	136 ms		
/mtscommerceservices/v2/mtsb2cSelfcare/users/current/info	498 ms	462 ms		
/mtscommerceservices/v2/mtsb2c/users/current/carts/<masked>/validate	133 ms	82.9 ms		
/mtscommerceservices/v2/mtsb2c/users/current/carts/<masked>	83.1 ms	44.4 ms		
/mtscommerceservices/v2/mtsb2c/users/anonymous/homeaddress/municipalities	72.3 ms	4.26 ms		
/mtscommerceservices/v2/mtsb2c/users/anonymous/carts	58.3 ms	15.7 ms		
/mtscommerceservices/v2/mtsb2c/users/current/carts	50.5 ms	13.5 ms		
/mtscommerceservices/v2/mtsb2cSelfcare/users/current	16.8 ms	8.41 ms		
/mtscommerceservices/v2/mtsb2c/users/current/addresses	11.5 ms	7.22 ms		
/mtscommerceservices/v2/mtsb2c/products/search NumberOfS...	10.7 ms	4.19 ms		
/mtscommerceservices/v2/mtsb2c/tariffs/mtsb2cProductCatalog/Online/mobiletariff	8.34 ms	6 ms		

According to the provided documentation, I/O operations seem to be the root cause of the slow responses of the login functionality:



- We recommend performing performance tests for the SSO Client to ensure that the integration can cope with the load coming from commerce.

6.3 Environment Sizing

Although SAP Commerce Cloud provides CPU based

<https://help.sap.com/docs/r/0fa6bcf4736c46f78c248512391eb467/latest/en-US/f65abe2e26864fecad4011dad03758c5.html>

on Storefront and API aspects, it is important to ensure the size of your production environment is adequately sized for your baseline traffic, including your solr/backoffice/backgroundProcessing aspects.

7 DATA MAINTENANCE

Data maintenance is an essential part of an SAP Commerce Solution implementation, and one must consider the impact of the growing amounts of data that is generated on the overall health of the system. There needs to be a strategy to identify what data is live, what can be archived and what can be removed to keep the system running at peak performance and avoiding unplanned slowdowns months after the initial go-live.

The following part of this Cloud Readiness Check will review the implementation for the recommended practices identified in [Data Maintenance and Cleanup](#).

7.1 Large Tables

Based on our analysis you should consider clean-up of the following tables:

Flexible Query

SQL Query

Search result

Execution statistics

History

Commit: OFF

Search:

Show

10

 entries

name	rows	reserved_kb	data_kb	index_kb	unused_kb
dbo.tmaabstractorderprice	5797892	6821792	5137696	1682880	1216
dbo.tasklogs	1417709	2454688	2389440	64256	992
dbo.products1sn	59242	836888	830928	5624	336
dbo.users4sn	15839	252632	250672	1704	256
dbo.addresses23sn	49776	202576	199280	3208	88
dbo.medias	67784	556480	145512	280144	130824
dbo.solrindexoperation	476274	168928	129408	38504	1016
dbo.props	84669	173520	128712	33032	11776
	106024	75840	69368	4848	1624
dbo.spp2tmaptrel	326152	156544	68848	85408	2288

Showing 1 to 10 of 200 entries

Previous

1

2

3

4

5

...

20

Next

7.2 CleanUp Cronjob

CleanUpCronJobStrategy can be used to clean up instances of the CronJob. This strategy is based on maintenance framework created for processing instances of any type. For more information, see [cronjob service](#). Default settings for the cron job cleanup mechanism is to remove instances of the CronJob, which fulfil the following:

- Has no trigger
- Are successfully finished
- Are older than 14 days

SAP recommends declaring and activate all clean up cronjob necessary to keep database platform in good shape.

Beware: the information here can be interesting for an audit trail. Make sure to backup the database and only cleanup if there is no need for these entries anymore.

If it has been a long time since you have cleaned up your cronjob history you may consider doing a [one-time cleanup](#).

Check the appendix for a list of jobs that should be cleaned up.

7.3 Sample Data

Often SAP Commerce project is started by using sample data which can help developer to accelerate the customization. At the end of the project, it is important to clean up this not relevant data to not disturb business team who will be in charge to maintain the system alive.

Make sure backoffice objects and business settings are relevant for the business

7.4 Update JSON

It is recommended to use an [Update JSON](#) to ensure that sample data will not be loaded into the system when a platform update is executed via Cloud Portal.

7.5 Generic Audit

The feature 'Generic Audit' has been introduced in 6.6 in order fulfil the GDPR issued by the European Union. The auditing tracks every persistence action (CRUD) of the specified types in a change log and is enabled by default (currently enabled). The change logs are stored in a dedicated table per type in the database (table name: <deploymentName><numericTypeCode>sn), which might grow very quickly.

Configuration Properties

apply all reset all

Show unsaved only: ☐ OFF

Show 10 entries Search: auditing

Key	Value
New key...	New value... add
auditing.blacklist	itemsynctimestamp.joblog.logfile.jobmedia.task,
auditing.enabled	false

Showing 1 to 2 of 2 entries (filtered from 2,962 total entries) Previous 1 Next

If you are not obliged to meet GDPR or other data protection regulations, disable this feature completely by setting the following property:

```
auditing.enabled=false
```

Otherwise, configure the auditing wisely by disabling logging for non critical types such as products (for more information see <https://help.sap.com/viewer/d0224eca81e249cb821f2cdf45a82ace/1905/en-US/3e2b8491449d43b391eb1ce9c73cbb7f.html>):

```
audit.product.enabled=false
```

Note: Configuring auditing via properties is possible on a type level only. If a finer level of granularity is required, use [Change Log Filtering](#).

SAP CX Field Services recommends to have a clean up strategy in place for types with configured generic audit. This can be done by either defining a clean up CronJob leveraging the provided interface (see [here](#)) or by a custom implementation.

7.6 Promotions

The promotions module of SAP Commerce Cloud internally uses a promotion engine that is entrusted to perform evaluations on cart and perform selected actions for eligible carts. The promotion engine is based on Drools engine. It is important to keep in mind that misuse of the promotion engine might cause performance issues.

The following articles will help you keep a clean set of promotions or to optimize your promotions for best performance:

- [Top 10 Recommendations for Improving the Performance of your Commerce Cloud Promotion Engine](#)
- [Maintenance Cron Job for Drools Rules](#)

7.7 Data Migration

During your go-live a data and media migration will likely be needed. Be it a copy from other SAP Commerce installation or initial load from other systems.

In case of migrations from an existing SAP Commerce installation. The following guidelines are available to assist with migrating database and media:

- [Migrating to SAP Commerce Cloud: Data and Media Migration](#)
- [Commerce Migration Toolkit](#)
- [Migrate to SAP Commerce Cloud: Migrate Media with AzCopy](#)

In case of initial load, or migration from other tools. The following can be leveraged:

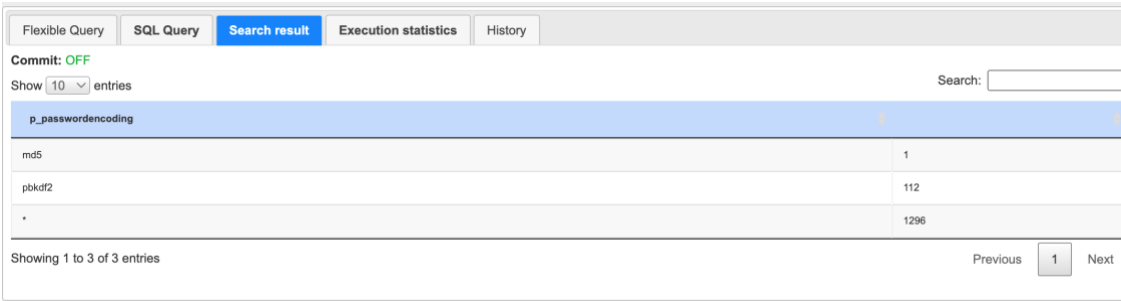
- [Strategies for Loading Data in SAP Commerce Cloud](#)

7.8 Other Maintenance Tasks

The above is non-exhaustive, and as for any IT system always ensure that a collection and rolling system in place for all log and troubleshooting data, as well as automated database and filesystem backup and archival procedures.

8 APPLICATION SECURITY REVIEW

Access to features within Commerce Cloud should be restricted to the appropriate users and sensitive data should be stored properly. We have reviewed several common issues found with user access and have outlined only those affecting your solution in the following table (checks that passed are not included):

Issue	Recommendation								
Transparent Attribute Encryption	Although the database is encrypted, you should consider not storing sensitive data (e.g. credit card number) in SAP Commerce Cloud and store it in the system where it is required. If it is required for SAP Commerce Cloud, ensure you are using Transparent Attribute Encryption (TAE) for these specific attributes, items: See documentation								
Password Encoding	Some passwords were found to not be encoded using the following flexible search SELECT {passwordencoding}, count(0) FROM {user} GROUP BY {passwordencoding}  The screenshot shows the 'Search result' tab of a Flexible Query. It displays a table with the following data: <table border="1"> <thead> <tr> <th>passwordencoding</th> <th>count(0)</th> </tr> </thead> <tbody> <tr> <td>md5</td> <td>1</td> </tr> <tr> <td>pbkdf2</td> <td>112</td> </tr> <tr> <td>*</td> <td>1296</td> </tr> </tbody> </table> Showing 1 to 3 of 3 entries	passwordencoding	count(0)	md5	1	pbkdf2	112	*	1296
passwordencoding	count(0)								
md5	1								
pbkdf2	112								
*	1296								
Default Users Credentials	There are 1296 passwords not encoded. See documentation We found some issues, please refer to the table below. However, since we did the check on S1, we highly recommend conducting the test in production to ensure that only necessary default users are used but also the corresponding encryption method is being used. To do it please refer to: https://help.sap.com/docs/SAP_COMMERCE_CLOUD_PUBLIC_CLOUD/ddc1ef9fc7fa48cb90f2b57c061a4012/07e594acc6544076b715407b14dfd0bb.html?version=v1905&locale=en-US								

Following table will give some more details:

Status	User	Password Encoding	Comment
✓	admin	pbkdf2	The sample user still exists but at least the password does not default.
✓	agent1	-	Does not exists. All fine.
✓	agent2	-	Does not exists. All fine.
✓	agent4	-	Does not exists. All fine.
⚠	anonymous	pbkdf2	Login disabled. Change empty password to the random one.
⚠	BackofficeIntegrationAdministrator	-	Login disabled. Change empty password to the random one.
⚠	BackofficeIntegrationAgent	-	Login disabled. Change empty password to the random one.
✓	BackofficeProductAdministrator	-	Does not exists. All fine.
✓	BackofficeProductManager	-	Does not exists. All fine.
✓	BackofficeWorkflowAdministrator	-	Does not exists. All fine.
✓	BackofficeWorkflowUser	-	Does not exists. All fine.
⚠	cmseditor	plain text	Login disabled. Change empty password to the random one.
⚠	cmsmanager	plain text	Login disabled. Change empty password to the random one.
✓	cmsmanager-apparel-de	-	Does not exists. All fine.
✓	cmsmanager-apparel-uk	-	Does not exists. All fine.
✓	cmsmanager-electronics	-	Does not exists. All fine.
✓	cmsmanager-powertools	-	Does not exists. All fine.
⚠	cmspublisher	plain text	Login disabled. Change empty password to the random one.
⚠	cmsreviewer	plain text	Login disabled. Change empty password to the random one.
⚠	cmstranslator	plain text	Login disabled. Change empty password to the random one.
⚠	cmstranslator-Annette	plain text	Login disabled. Change empty password to the random one.
⚠	cmstranslator-Seb	plain text	Login disabled. Change empty password to the random one.
✓	csagent	-	Does not exists. All fine.
✓	cxmanager	-	Does not exists. All fine.
✓	cxuser	-	Does not exists. All fine.

Status	User	Password Encoding	Comment
⚠	hac_editor	pbkdf2	Login disabled. Change default password to the random one.
⚠	hac_viewer	pbkdf2	Login disabled. Change default password to the random one.
✓	integration-admin	-	Does not exists. All fine.
✓	integrationmonitoringtestuser	-	Does not exists. All fine.
✓	integrationservicestandarduser	-	Does not exists. All fine.
✓	integrationservicetestuser	-	Does not exists. All fine.
✓	marketingmanager	-	Does not exists. All fine.
✓	men@hybris.com	-	Does not exists. All fine.
✓	menover30@hybris.com	-	Does not exists. All fine.
✓	menvipbronze@hybris.com	-	Does not exists. All fine.
✓	menvipgold@hybris.com	-	Does not exists. All fine.
✓	productmanager	-	Does not exists. All fine.
✓	salesrep1@test.com	-	Does not exists. All fine.
✓	salesrep2@test.com	-	Does not exists. All fine.
✓	salesrep3@test.com	-	Does not exists. All fine.
✓	sapInboundB2BCustomerUser	-	Does not exists. All fine.
✓	sapInboundB2CCustomerUser	-	Does not exists. All fine.
✓	sapInboundClassificationUser	-	Does not exists. All fine.
✓	sapInboundOMMOrderUser	-	Does not exists. All fine.
✓	sapInboundOMSOrderUser	-	Does not exists. All fine.
✓	sapInboundPriceUser	-	Does not exists. All fine.
✓	sapInboundProductUser	-	Does not exists. All fine.
✓	sapInboundUser	-	Does not exists. All fine.
⚠	searchmanager	pbkdf2	Login disabled. Change empty password to the random one.
✓	sellerapprover	-	Does not exists. All fine.
✓	Test-Employee	-	Does not exists. All fine.
✓	testcmsmanager	-	Does not exists. All fine.
✓	vipbronze@hybris.com	-	Does not exists. All fine.
✓	vipgold@hybris.com	-	Does not exists. All fine.
✓	vipsilver@hybris.com	-	Does not exists. All fine.
✓	WarehouseAdministrator	-	Does not exists. All fine.
✓	WarehouseAgent	-	Does not exists. All fine.

Status	User	Password Encoding	Comment
✓	WarehouseManager	-	Does not exists. All fine.
✓	wfl_marketing	-	Does not exists. All fine.
✓	wfl_marketing_DE	-	Does not exists. All fine.
✓	wfl_marketing_EN	-	Does not exists. All fine.
✓	wfl_marketing_ES	-	Does not exists. All fine.
✓	wfl_marketing_FR	-	Does not exists. All fine.
✓	wfl_marketing_GB	-	Does not exists. All fine.
✓	wfl_marketing_GSW	-	Does not exists. All fine.
✓	wfl_marketing_IT	-	Does not exists. All fine.
✓	wfl_marketing_SWE	-	Does not exists. All fine.
✓	wfl_productManagement	-	Does not exists. All fine.
✓	wfl_purchase	-	Does not exists. All fine.
✓	wfl_translator_DE	-	Does not exists. All fine.
✓	wfl_translator_EN	-	Does not exists. All fine.
✓	wfl_translator_ES	-	Does not exists. All fine.
✓	wfl_translator_FR	-	Does not exists. All fine.
✓	wfl_translator_GB	-	Does not exists. All fine.
✓	wfl_translator_GSW	-	Does not exists. All fine.
✓	wfl_translator_IT	-	Does not exists. All fine.
✓	wfl_translator_SWE	-	Does not exists. All fine.
✓	women@hybris.com	-	Does not exists. All fine.
✓	womenvipgold@hybris.com	-	Does not exists. All fine.
✓	womenvipsilver@hybris.com	-	Does not exists. All fine.
✓	yacctestcsagent	-	Does not exists. All fine.
✓	priceimport	pbkdf2	Does not exists. All fine.
✓	qa_commercemanager	-	Does not exists. All fine.
✓	qa_customerservicemanager	-	Does not exists. All fine.
✓	qa_customerserviceagent	-	Does not exists. All fine.
✓	qa_productcontenteditor	-	Does not exists. All fine.
✓	qa_technicalsupport	-	Does not exists. All fine.
✓	qa_financemanager	-	Does not exists. All fine.
✓	qa_categorymanager	-	Does not exists. All fine.
✓	qa_priceadmin	-	Does not exists. All fine.
✓	qa_systemadmin	-	Does not exists. All fine.

Status	User	Password Encoding	Comment
✓	unassignedEmployee	-	Does not exists. All fine.
✓	northAmericaEmployee	-	Does not exists. All fine.
✓	usaEmployee	-	Does not exists. All fine.
✓	canadaEmployee	-	Does not exists. All fine.
✓	canadaAdmin	-	Does not exists. All fine.
✓	globalAdmin	-	Does not exists. All fine.
✓	usaAdmin	-	Does not exists. All fine.
✓	usaManager	-	Does not exists. All fine.

9 FURTHER INFO

SAP is here to support you during and after your go-live. Here is a list of possible services included with your subscription that you may choose to use.

Deployment Readiness

If you require an additional review of your system around your go-live period, we recommend you schedule a Deployment Readiness Continuous Quality Check from SAP Enterprise Support. Please find more details about this service [here](#).

Performance Health Check

The Performance Health Check service is comprised of periodic analysis of your SAP Commerce Cloud Production storefront to identify application hotspots and top website response time contributors. The aim of this service is to highlight areas of the SAP Commerce Cloud platform which may be operating with degraded service due to incorrect configuration or customization. Consult with your Customer Engagement Executive or Account Executive to determine how often you qualify for this service.

Value Maximization Assessment

The Value Maximization Assessment, which is available only to professional and enterprise customers, provides a one-time analysis of your individual adoption of SAP Commerce Cloud's native features and functionalities after you've gone live and compares them with recommended practices to provide recommendations for increasing the value realized via your digital commerce platform. This analysis will result in an assessment report on key areas of focus, an overall adoption scorecard, and recommendations to drive more adoption. The findings will be documented, delivered via real-time presentation, and result in a jointly defined ongoing adoption plan. Consult with your Customer Engagement Executive or Account Executive to determine how to use this service.

Go-Live Checklist

In addition, you may find some of the following articles helpful:

- [Checklist for Migrating to SAP Commerce Cloud in the Public Cloud](#)
- [Go-live Readiness for SAP Commerce Cloud](#)
- [Managing Performance in your SAP Commerce Cloud Project](#)
- [Using Dynatrace to Optimize the Performance of your SAP Commerce Cloud solution](#)
- [Implement Continuous Delivery with SAP Commerce Cloud](#)
- [Data Maintenance and Cleanup](#)
- [User Management](#)
- [Strategies for Loading Data into SAP Commerce Cloud](#)

10 APPENDIX

10.1 Cronjobs to be removed

For readability purposes, the table below displays only first 100 cronjobs.

Code	Job definition	Current status	Last result	Timetable
0000ELGT	sync mtsB2CProductCatalog:Stage d->Online	NEW	N/A	Not scheduled
0000ELGS	sync mtsB2CProductCatalog:Stage d->Online	NEW	N/A	Not scheduled
0000ELGR	sync mtsB2CProductCatalog:Stage d->Online	NEW	N/A	Not scheduled
0000ELGQ	sync mtsB2CProductCatalog:Stage d->Online	NEW	N/A	Not scheduled
0000ELGP	sync mtsB2CProductCatalog:Stage d->Online	NEW	N/A	Not scheduled
0000ELGO	sync mtsB2CProductCatalog:Stage d->Online	NEW	N/A	Not scheduled
0000EKOW	sync mtsB2CProductCatalog:Stage d->Online	RUNNING	N/A	Not scheduled
0000EJZR	ImpEx-Import	FINISHED	SYSTEM ERROR	Not scheduled
0000EJYE-ImpEx-Import	ImpEx-Import	FINISHED	SYSTEM ERROR	Not scheduled
0000EJYD-ImpEx-Import	ImpEx-Import	FINISHED	SYSTEM ERROR	Not scheduled
0000EJXK-ImpEx-Import	ImpEx-Import	FINISHED	SYSTEM ERROR	Not scheduled
0000EJ5W	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5V	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5U	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5T	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled

0000EJ5S	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5R	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5Q	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5P	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5O	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5N	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5M	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5L	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5K	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5J	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5I	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5H	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5G	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5F	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5E	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5D	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EJ5C	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled

updateCustomerDetailsJob-9a1bee4a-dff4-4371-a269-9ebd86c03cac_aleksandar-aca123@hotmail.com-93007f28-128d-4d84-aff5-accd1c67739f	updateCustomerDetailsJob	NEW	N/A	Not scheduled
0000EIG7	ImpEx-Import	FINISHED	SYSTEM ERROR	Not scheduled
0000EIEU-ImpEx-Import	ImpEx-Import	FINISHED	SYSTEM ERROR	Not scheduled
0000EIET-ImpEx-Import	ImpEx-Import	FINISHED	SYSTEM ERROR	Not scheduled
0000EIE0-ImpEx-Import	ImpEx-Import	FINISHED	SYSTEM ERROR	Not scheduled
0000EHMK	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHMJ	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHMI	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHMH	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHMG	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHMF	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHME	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHMD	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHMC	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHMB	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHMA	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHM9	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled

0000EHM8	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHM7	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHM6	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHM5	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHM4	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHM3	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHM2	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHM1	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHM0	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHLZ	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHL Y	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHLX	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHLW	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHLV	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHLU	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHLT	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EHLS	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled

0000EGVG	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGVF	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGVE	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGVD	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGVC	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGVB	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGVA	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGV9	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGV8	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGV7	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGV6	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGV5	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGV4	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGV3	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGV2	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGV1	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGV0	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled

0000EGUZ	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUY	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUX	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUW	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUV	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUU	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUT	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUS	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUR	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUQ	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUP	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUO	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUN	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUM	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUL	sync mtsB2CProductCatalog:Stage d->Online	FINISHED	SUCCESS	Not scheduled
0000EGUK	solrIndexerJob	FINISHED	SUCCESS	Not scheduled
0000EGUJ	solrIndexerJob	FINISHED	SUCCESS	Not scheduled

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